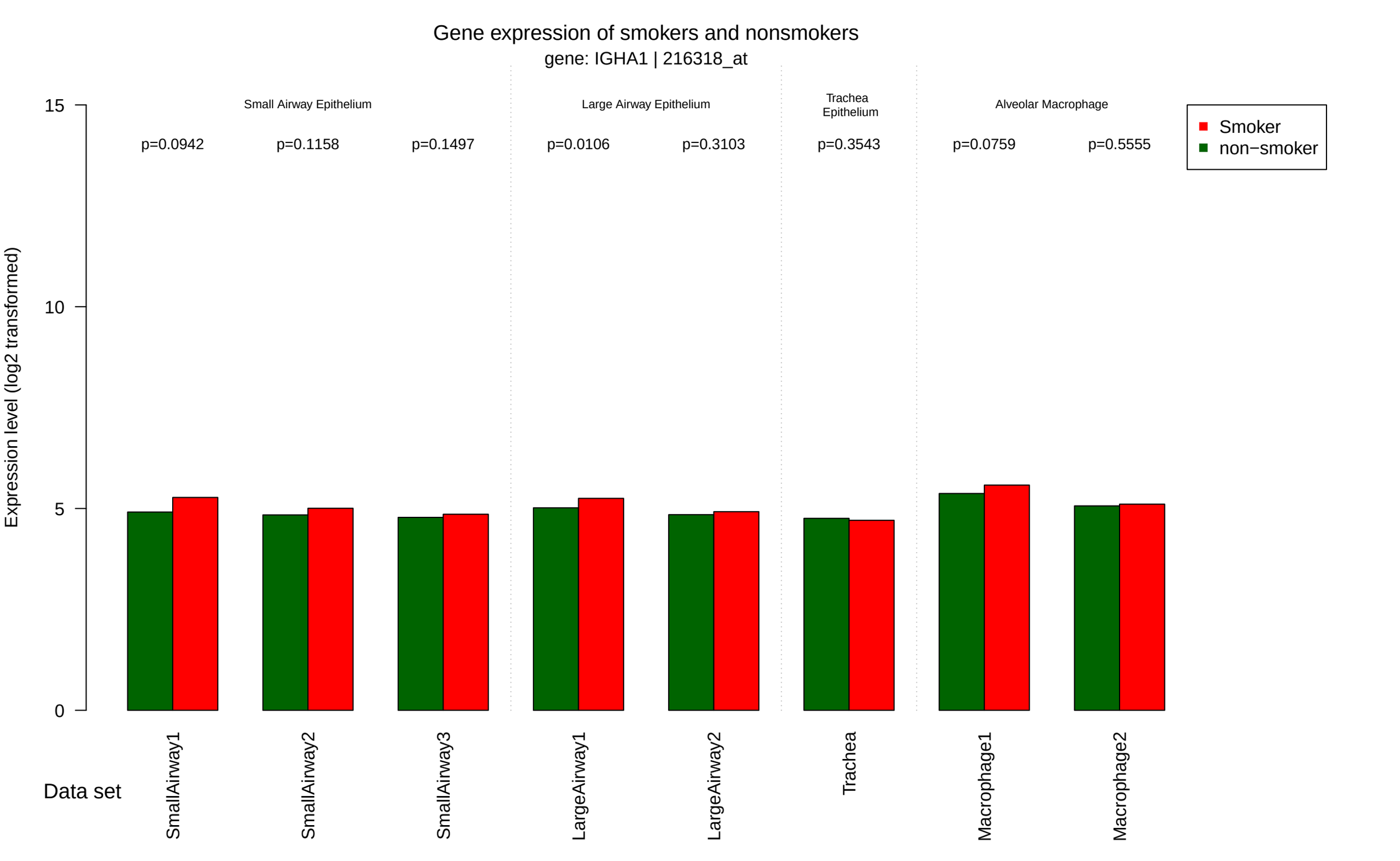
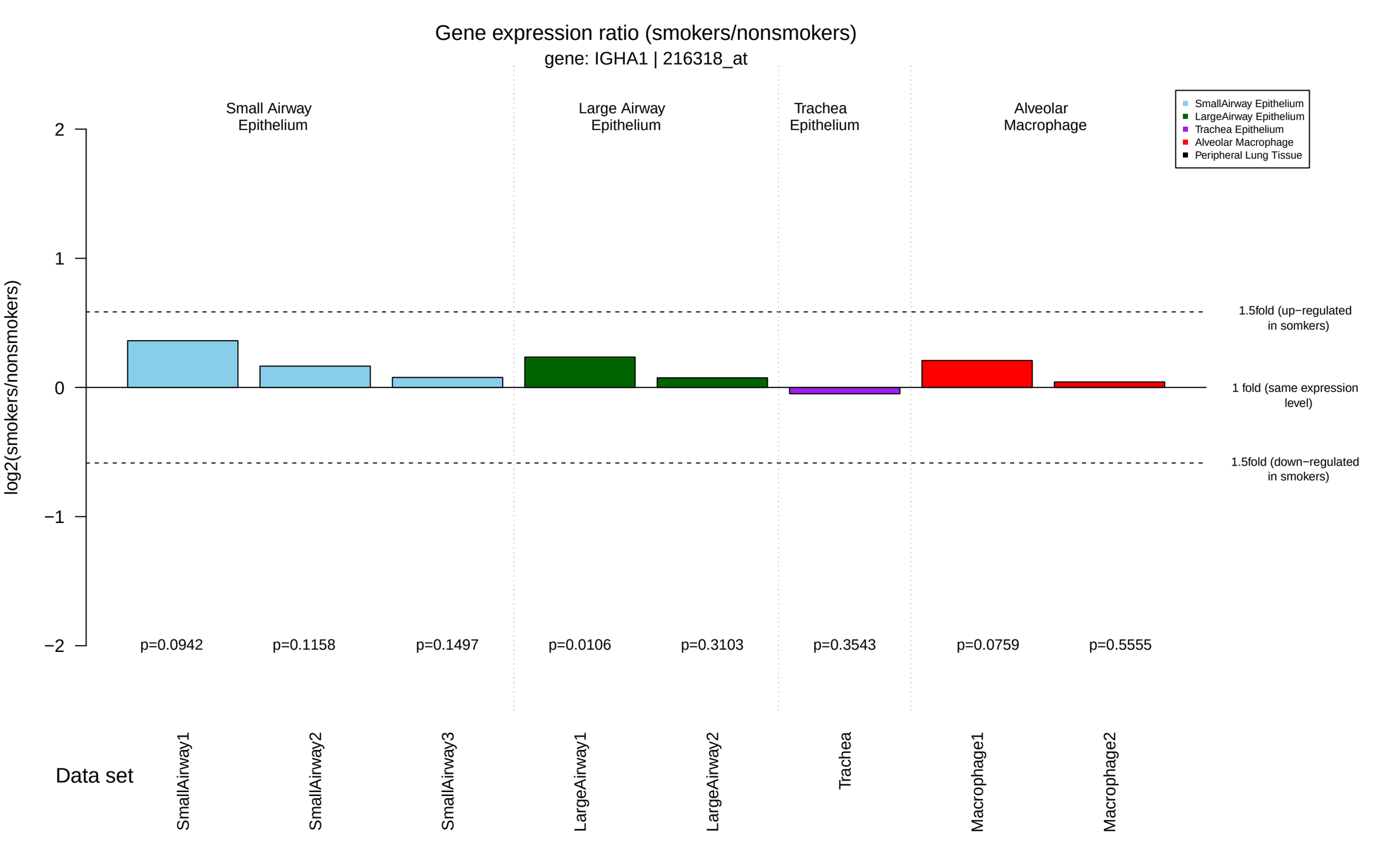
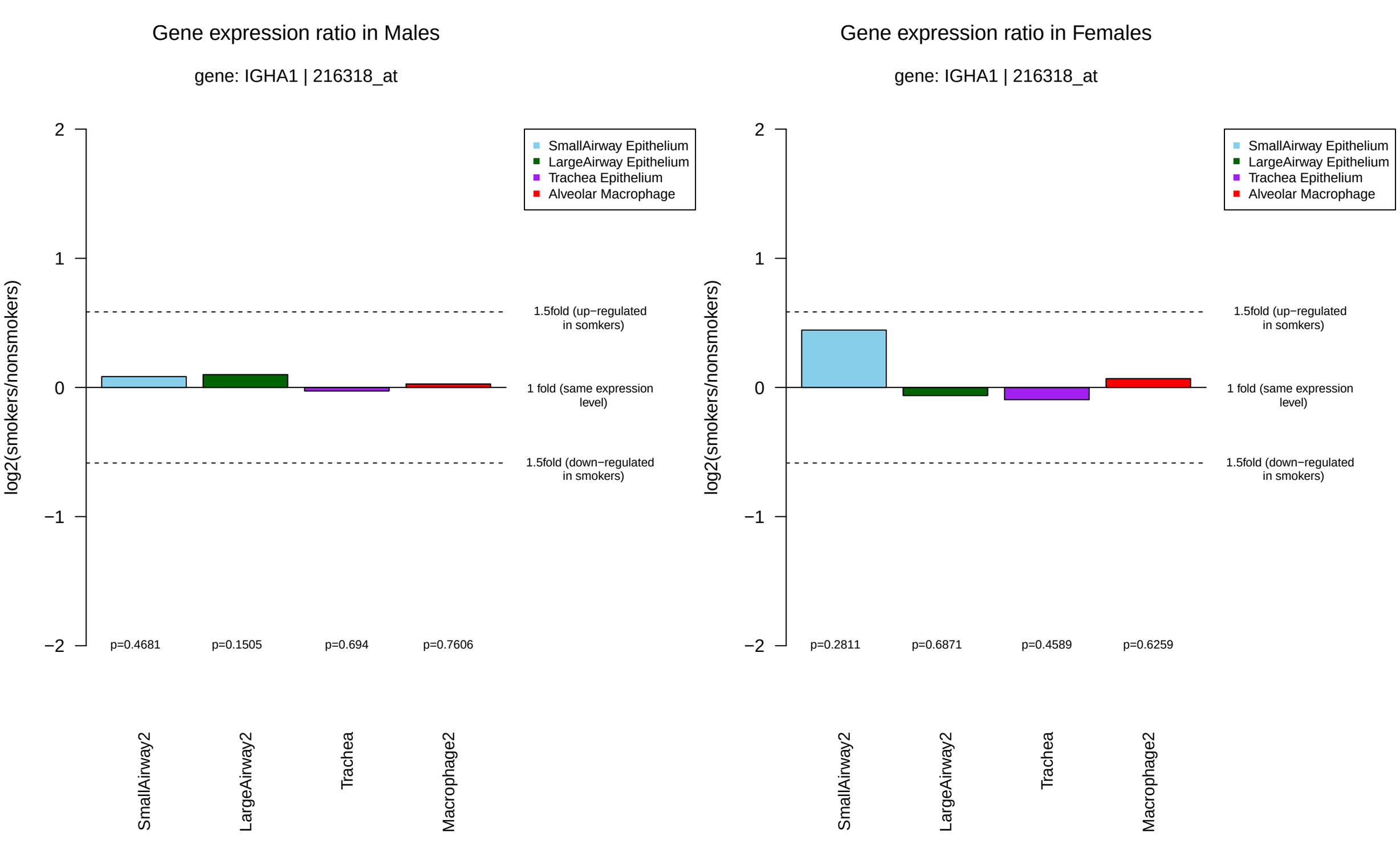
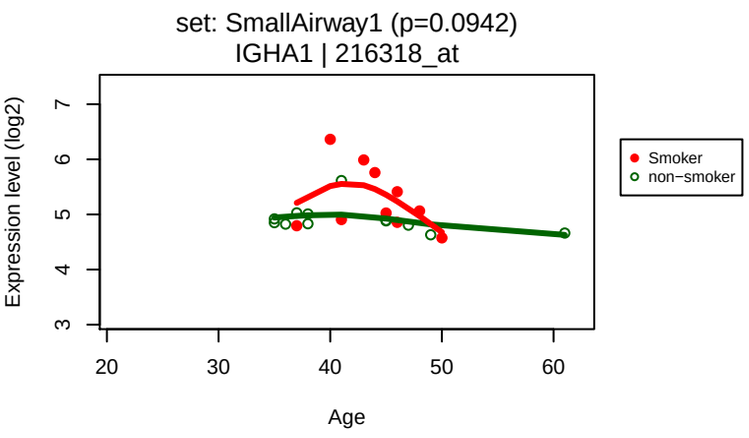
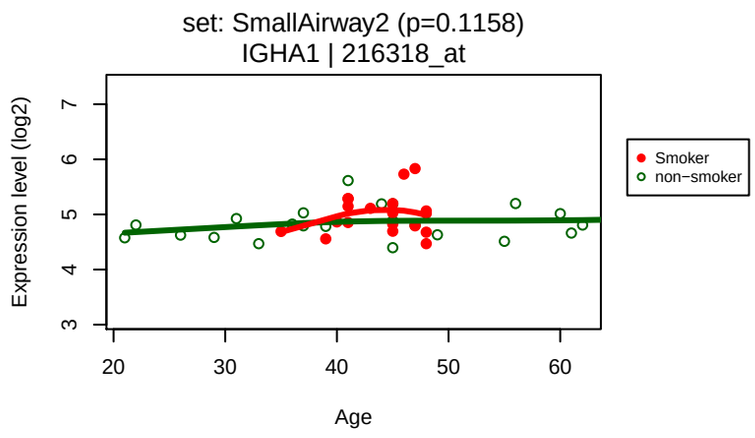
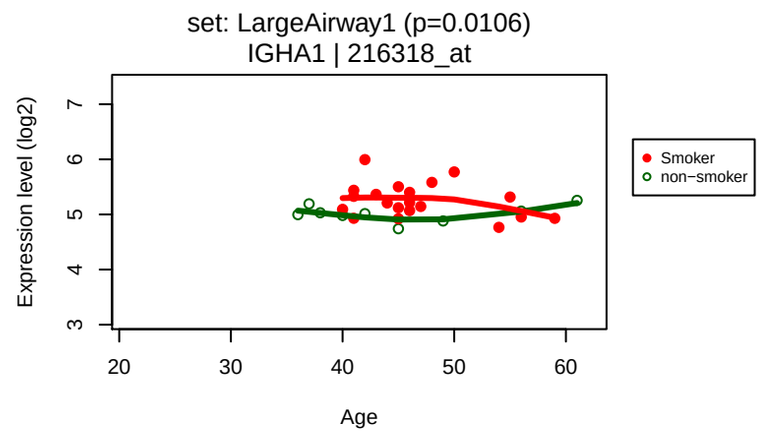

Gene expression of smokers and nonsmokers
gene: IGHA1 | 216318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0942
p=0.1158
p=0.1497
p=0.0106
p=0.3103
p=0.3543
p=0.0759
p=0.5555
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

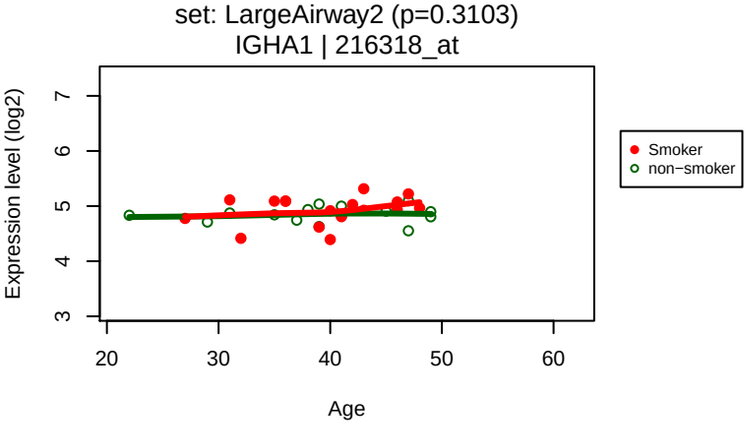

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 | 216318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0942
p=0.1158
p=0.1497
p=0.0106
p=0.3103
p=0.3543
p=0.0759
p=0.5555
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

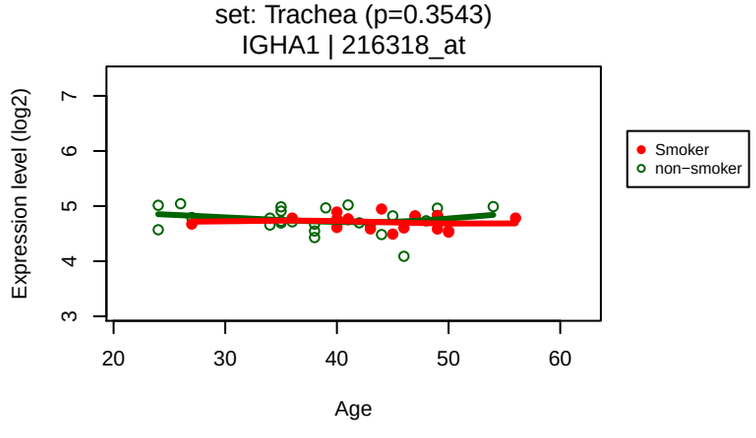

Gene expression ratio in Males
gene: IGHA1 | 216318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4681
p=0.1505
p=0.694
p=0.7606
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IGHA1 | 216318_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2811
p=0.6871
p=0.4589
p=0.6259
SmallAirway2
LargeAirway2
Trachea
Macrophage2

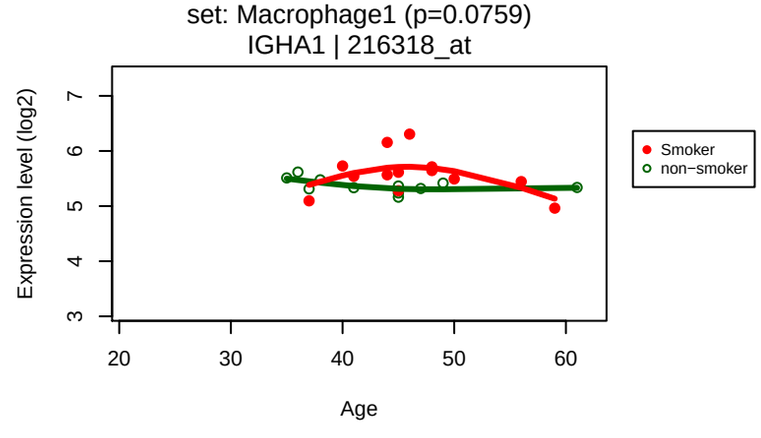

set: SmallAirway1 (p=0.0942)
IGHA1 | 216318_at
Expression level (log2)
Age
Smoker
non−smoker

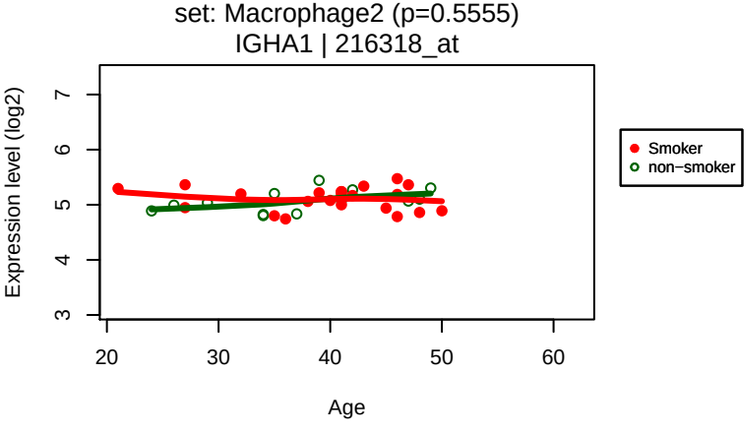

set: SmallAirway2 (p=0.1158)
IGHA1 | 216318_at
Expression level (log2)
Age
Smoker
non−smoker

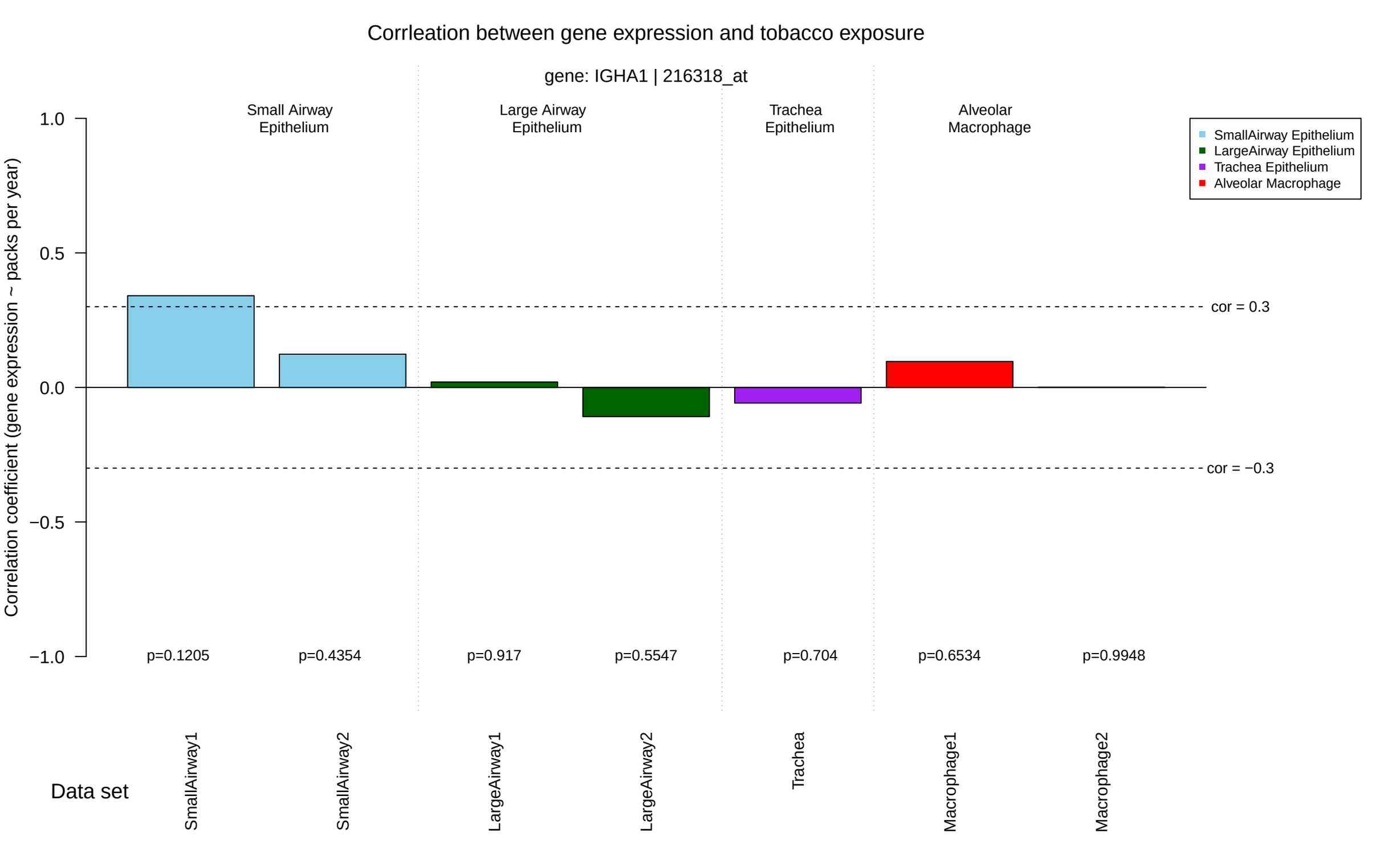

set: LargeAirway1 (p=0.0106)
IGHA1 | 216318_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3103)
IGHA1 | 216318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3543)
IGHA1 | 216318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0759)
IGHA1 | 216318_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5555)
IGHA1 | 216318_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 | 216318_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1205
p=0.4354
p=0.917
p=0.5547
p=0.704
p=0.6534
p=0.9948
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2